An Architecture for YANG-Push to Message Broker Integration

draft-ietf-nmop-yang-message-broker-integration-11

Motivation and architecture of a native
YANG-Push notifications and YANG Schema integration
into Message Broker and YANG Schema Registry

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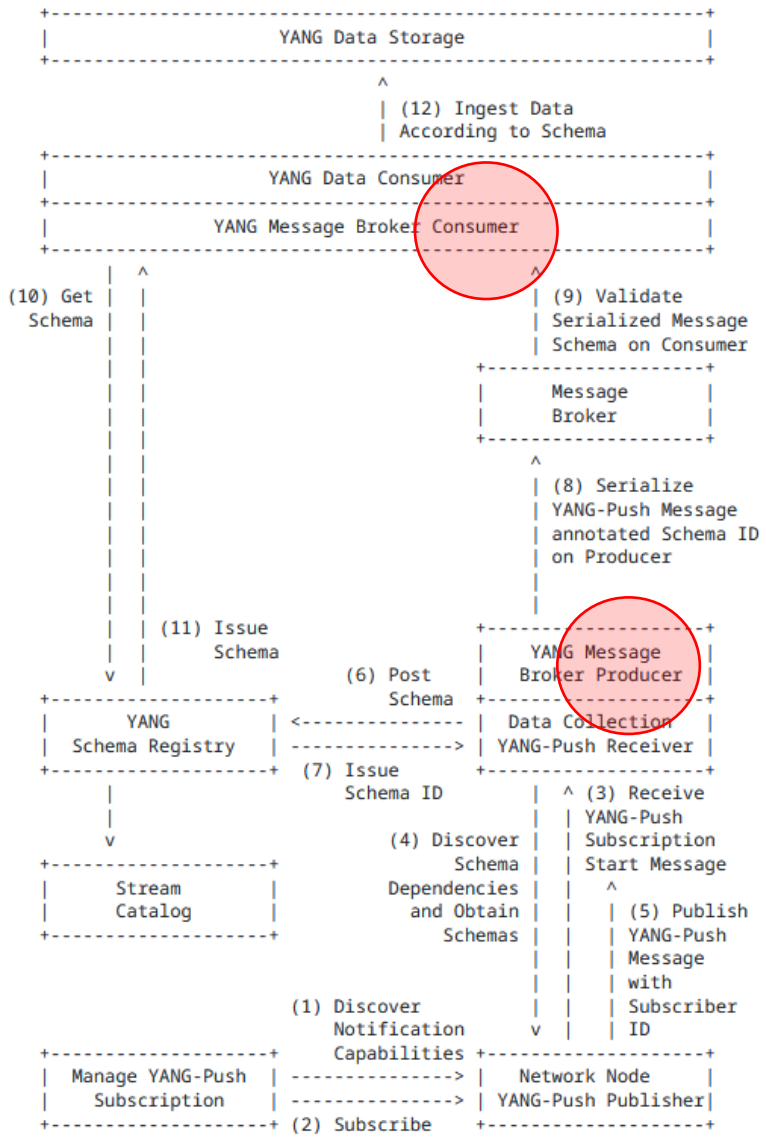
Status from -10 and -11 and Next Steps

Status

- Expanded system architecture picture with YANG Message Broker Producer and Consumer
- Expanded terminology section with terms defined by the document.
- Expanded YANG-Push Receiver and YANG Schema Registry to describe and example how YANG features are being covered in schema registration.
- Added IETF 125 in the milestone appendix to describe the current implementation states and the planned hackathon items.
- Minor editorial edits.

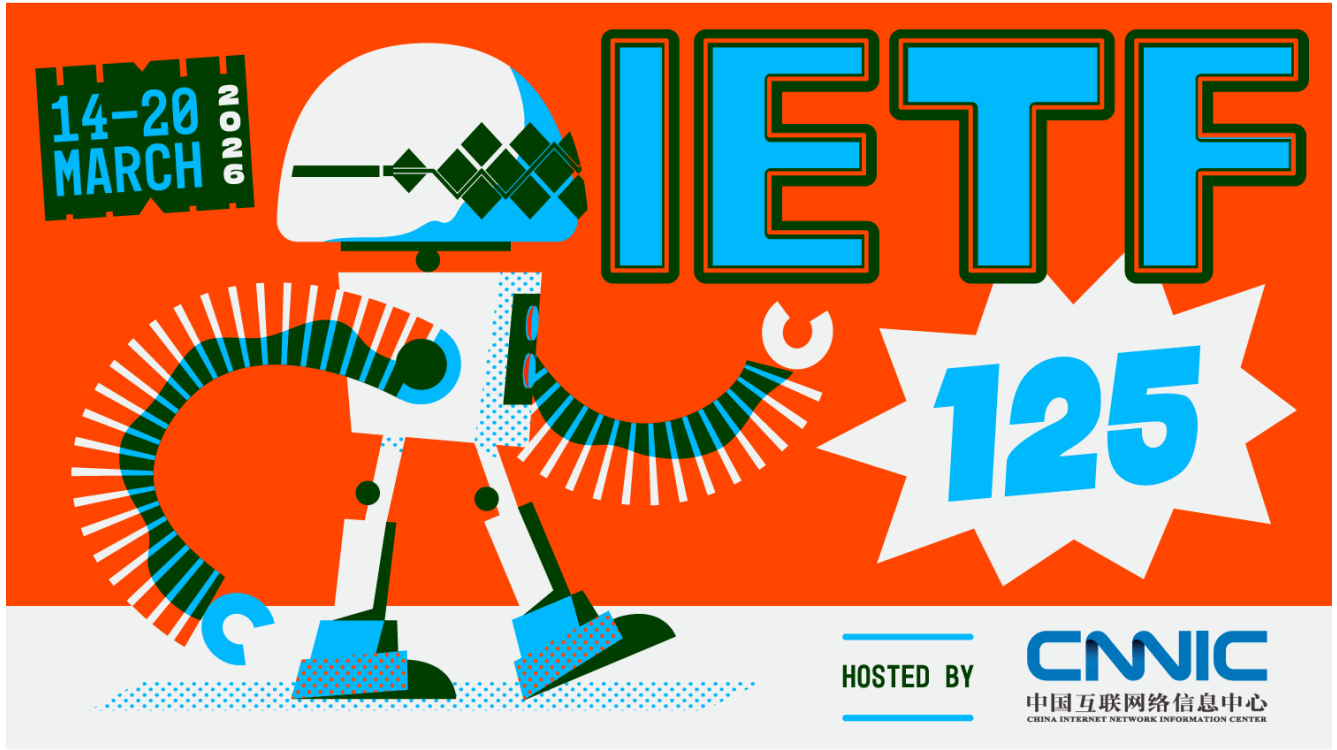
Next Steps

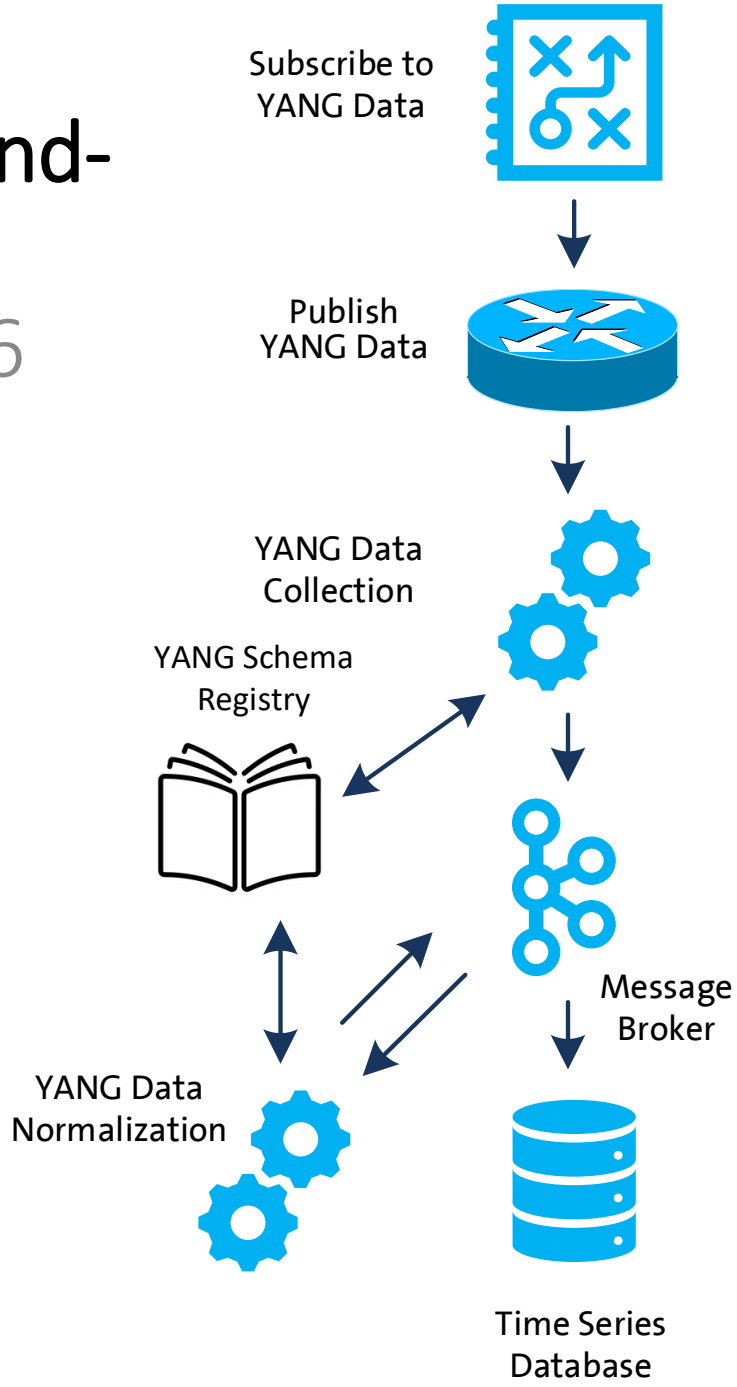
➢ Add security and operations considerations section.



Validate YANG-Push to Message Broker End-To-End Data Processing Chain

IETF 125 Hackathon, March 14-15th 2026





NMOP & NETCONF WG

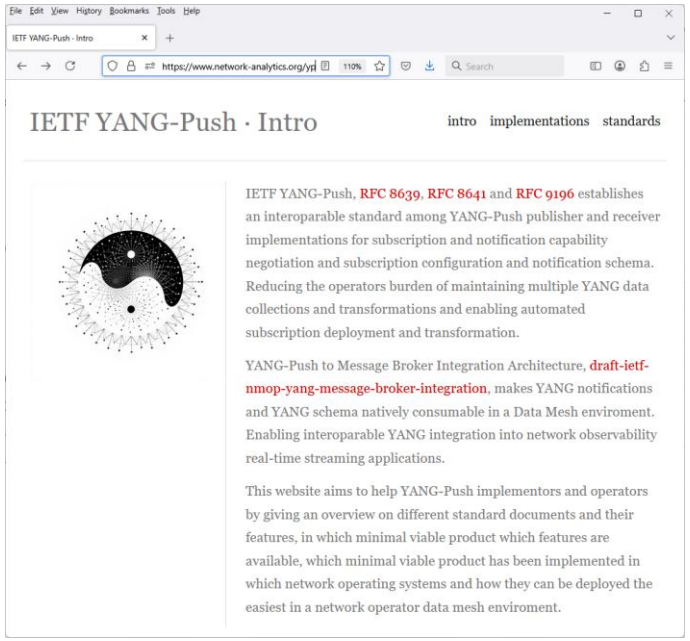
Hackathon Plan, Software and Website

Test Plan

Validate and verify

- 5 YANG-Push Publishers
- 2 YANG-Push Receivers
- 2 YANG-Push Network Telemetry Message
- 1 YANG Message Broker Producer and Schema Registry
- 3 YANG Message Broker Consumers

implementation in YANG data integration automation. Subscribe to YANG data on YANG-Publisher, obtain and register all YANG modules necessary to build YANG schema tree, register YANG schemas to Schema Registry and verify YANG notifications against scheme trees and produce and consume from Message Broker.



Software

https://www.network-analytics.org/yp/how-to-deploy.html

- YANG-Push Publisher - Cisco IOS XR, 6WIND VSR, Huawei NE (Router) and MA (OLT) and Arrcus Arcos
- YANG-Push Receiver – Netgauze and Pmacct
- YANG Message Broker Producer – Netgauze
- YANG Message Broker Consumer – Netgauze, Ciena Blueplanet UAA Workflow Engine, Cisco Crosswork Ingestion Service
- YANG Schema Registry – Apache Kafka YANG Plugin
- Libyang - https://github.com/CESNET/libyang/compare/master...devel
- Yangkit - https://github.com/yang-central/yangkit/pull/41
- Wireshark - udp-notif dissector

Hackathon – Repositories

Test Results on End to End Tests

- Netconf RPCs and YANG-Push JSON encoded notifications with wire captures from receiver
- YANG-Push Receiver cache content generation with YANG library context, YANG modules, metadata and notification schema validation logs
- YANG Message Broker Producer serialization logs
- YANG Schema Registry schema examples
- YANG Message Broker Consumer, serialization logs, YANG library context, YANG modules and message schema validation

ietf-network-analytics-document-status / 125 / Hackathon / Add file ···

graf3net Cisco Crosswork Ingestion Service d53f11b · last month History

Name	Last commit message	Last commit date
..		
messagebroker-consumer	Cisco Crosswork Ingestion Service	last month
messagebroker-producer/netgauze-prod...	testcases updated	last month
yangpush-publisher	testcases updated	last month
yangpush-receiver/netgauze-receiver	nit	last month
how-to-install-netgauze.md	how to install netgauze	last month
readme.md	updated	last month

readme.md

YANG-Push and Message Broker Architecture

As described in https://datatracker.ietf.org/doc/html/draft-ietf-nmop-yang-message-broker-integration-10#section-4

https://github.com/network-analytics/ietf-network-analytics-document-status/tree/main/125/Hackathon/

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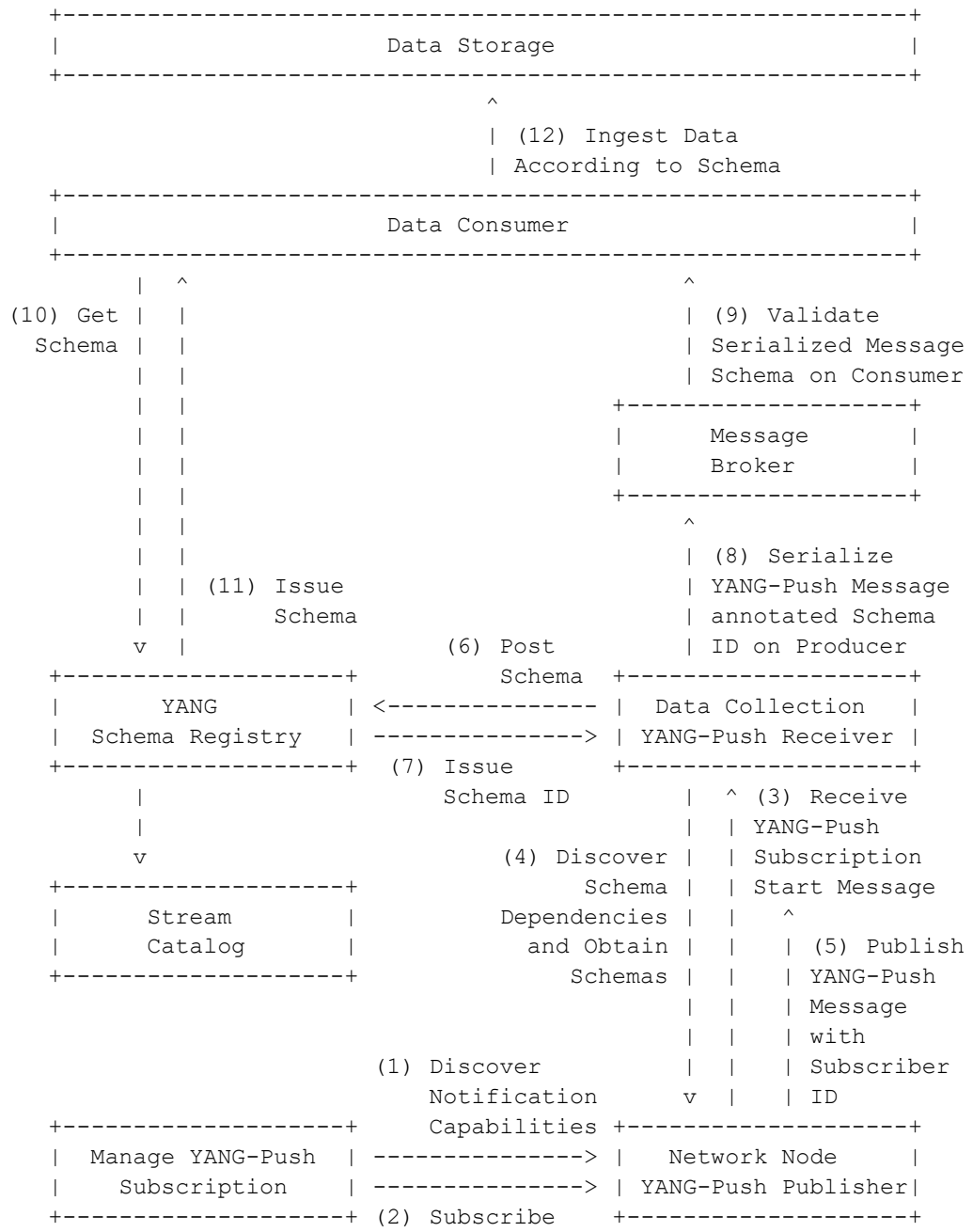
draft-ietf-nmop-yang-message-broker-integration
draft-ietf-nmop-message-broker-telemetry-message
draft-ietf-nmop-yang-message-broker-message-key
draft-netana-nmop-message-broker-bmp-telemetry-message

- Subscription to YANG Notifications
  RFC 8639
- Subscription to YANG Notifications for Datastore Updates
  RFC 8641
- UDP-based Transport for Configured Subscriptions
  draft-ietf-netconf-udp-notif
- Subscription to Distributed Notifications
  draft-ietf-netconf-distributed-notif
- Extensible YANG Model for YANG-Push Notifications
  draft-ietf-netconf-notif-envelope
- Support of Versioning in YANG Notifications Subscription
  draft-ietf-netconf-yang-notifications-versioning
- YANG Modules Describing Capabilities for Systems and Datastore Update Notifications
  RFC 9196
- YANG Notification Transport Capabilities
  draft-ietf-netconf-yp-transport-capabilities
- YANG Library
  RFC 8525
- Augmented-by Addition into the IETF-YANG-Library
  draft-ietf-netconf-yang-library-augmentation
- Validating anydata in YANG Library context
  draft-ietf-netmod-yang-anydata-validation
- Encoding of Data Modeled with YANG in the CBOR
  RFC 9254



YANG-Push

IETF 125 Publisher Implementation Status

	6WIND VSR	Huawei NE	Huawei MA	Cisco IOS XR	Arrcus Arcos	Open-Source
RFC 8639 YANG-Push Subscription	✓	✓	✓	✓	✓	
RFC 8641 YANG-Push Notification	✓	✓	✓	✓	✓	
draft-ietf-netconf-udp-notif	✓	✓	✓	✓	✓	✓
draft-ietf-netconf-distributed-notif	✓	✓	✓			
draft-ietf-netconf-notif-envelope	✓	✓	✓	✓	✓	
draft-ietf-netconf-yang-notifications-versioning	✓	✓	✓	✓		
RFC 8525 YANG Library	✓	✓	✓	✓		
draft-ietf-netconf-yang-library-augmentation	✓	✓	✓	✓		✓
RFC 9196 System and Notification Capabilities			✓	P		
draft-netana-netconf-yp-transport-capabilities			✓	✓	✓	
RFC 9254 CBOR Named Identifiers	✓					

NetGauze
Pmacct
udp-notif-c-collector
yang-library-augmentedby

Green marked describes new capability at IETF 125. "P" to partially implemented.

Operational IETF/IEEE YANG

IETF 125 Module Implementation Status

	6WIND VSR	Huawei NE	Huawei MA	Cisco IOS XR	Arrcus Arcos
ietf-interfaces.yang (inventory, state and statistic)	✓	P	✓		
ietf-hardware.yang (inventory, state and statistic)	✓		✓		
ietf-alarms.yang (state)	✓		✓		
ieee802-dot1ab-lldp.yang (inventory and state)					
ieee802-dot1ax.yang (inventory, state and statistic)					
ietf-bfd-ip-sh.yang (state)					
ietf-bfd-ip-mh.yang (state)					
ietf-bfd-lag.yang (state)					
ietf-isis.yang (state)					

Green marked describes new capability at IETF 125. "P" to partially implemented.

YANG Message Broker

IETF 125 Implementation Status

	NetGauze	Pmacct	Apache Kafka	Ciena Blueplanet	Cisco Crosswork
draft-ietf-nmop-message-broker-telemetry-message	✓	✓			
YANG-Push Receiver	✓	✓			
YANG Schema Retrieval	✓				
YANG Message Broker Producer	✓				
YANG Schema Registry			✓		
YANG Message Broker Consumer	✓			✓	✓
Validate Telemetry Message against YANG Schema	✓			✓	✓
draft-ietf-netmod-yang-anydata-validation	✓				
YANG Schema Comparison			✓		

NetGauze
Pmacct

Green marked describes new capability at IETF 125. "P" to partially implemented.

Thanks to…

- Paolo Lucente – Pmacct
- Scott Huan - Cisco
- Rob Wilton – Cisco
- Dan Voyer – Cisco
- Sivakumar Sundaravadivel – Ciena
- IJsbrand Wijnands – Arrcus
- Irfan Mohammad – Arrcus
- Benoit Claise - Everything OPS SRL
- Wanting Du – Swisscom
- Rafael Julio – Swisscom
- Leonardo Rodoni - Swisscom
- Ahmed Elhassany – Swisscom
- Thomas Graf – Swisscom

Backup

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Normative References, State and Reviewers

Working Group	Document Name	Document State	Reviews
NETCONF	draft-ietf-netconf-notif-envelope-04	WG Document (passed working group last call) and is looking for a shepherd	OPSDIR Early review by Joe Clarke YANGDOCTORS Early review (of -02) by Jürgen Schönwälder
NETCONF	draft-ietf-netconf-yang-library-augmentedby-17	RFC editor queue	OPSDIR Early review (of -08 and -14) by Sheng Jiang, Xiao Min YANGDOCTORS Early review (of -07 and -13) by Andy Bierman SECDIR review (of -14 and -15) by Rich Salz
NETCONF	draft-ietf-netconf-yang-notifications-versioning-11	Working group last call past- Pending shepherd review.	YANGDOCTORS Early Review due 2025-10-29 OPSDIR review (of -10) by Gabriele Galimberti
NETCONF	draft-ietf-netconf-yp-transport-capabilities-05	Submitted to IESG for Publication	OPSDIR IETF Last Call review by Xiao Min TSVART IETF Last Call review by Tommy Pauly YANGDOCTORS IETF Last Call review by Jan Lindblad OPSDIR Early review (of -03) by Xiao Min YANGDOCTORS Early review (of -01) by Jan Lindblad SECDIR IETF Last Call Review due 2025-10-18
NMOP	draft-ietf-nmop-yang-message-broker-integration-11	WG Document (request for working group last call at IETF 124)	
NMOP	draft-ietf-nmop-message-broker-telemetry-message-04	WG Document (request for working group last call at IETF 125)	YANGDOCTORS Early review (of -02) by Martin Björklund

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Informative References, State and Reviewers

Working Group	Document Name	Document State	Reviews
NETCONF	draft-ietf-netconf-udp-notif-25	AD review received. Pending confirmation.	OPSDIR Early review (of -21) by Tina Tsou YANGDOCTORS Early review (of -20) by Jürgen Schönwälder TSVART Early review (of -11) by Michael Tüxen TSVART Early Review due 2025-10-23
NETCONF	draft-ietf-netconf-udp-client-server-10	Publication Requested	OPSDIR Early review (of -07) by Ran Chen TSVART Early review (of -06) by Joerg Ott YANGDOCTORS Early review (of -06) by Jürgen Schönwälder GENART review (of -09) by Elwin Davies DNSDIR review (of -09) by Peter Van Dijk
NETCONF	draft-ietf-netconf-distributed-notif-18	AD review received. Pending confirmation.	OPSDIR Early review (of -14) by Jürgen Schönwälder YANGDOCTORS Early review (of -13 and -17) by Martin Björklund
NETMOD	draft-ietf-nmop-yang-anydata-validation-00	Working group adopted at IETF 124.	